

Sentiment Analysis with Hierarchical Reasoning Models and Mixture-of-Experts: Implementation and Repository Evaluation

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Abstract—This work implements and evaluates a heterogeneous *mixture-of-experts* framework for sentiment analysis that combines Hierarchical Reasoning Models (HRMs), classical machine-learning experts (TF-IDF with logistic regression or linear SVM), and deep learning / transformer encoders. Predictions are fused through stacking with five-fold out-of-fold meta-learning and through dense or sparse MoE gating. The thesis-scale protocol covers four public corpora (Sentiment140, IMDB, Amazon reviews, TweetEval Feminism), 35 trained model configurations, and 140 model–dataset pairs. The strongest stacking ensemble (STACK1) reaches macro-F1 0.8911 on IMDB and 0.8746 on Amazon under the reported settings. Across all conditions, traditional ML achieves the highest *mean* macro-F1 (0.6389), while transformers and HRMs average 0.3099 and 0.2502 respectively in this checkout—motivating ensemble design, imbalance-aware training, and architecture refinements. This paper also documents the live project tree: merged Parquet alignment, exported model and path inventories, regenerated charts, and a small HRAST per-stem evaluation slice for regression testing (not a second full benchmark). The emphasis is modular pipelines, reproducibility, and interpretability hooks rather than a new single-backbone classifier.

Index Terms—Sentiment analysis, mixture-of-experts, ensemble learning, hierarchical reasoning models, stacking, gating networks, natural language processing, macro-F1.

I. INTRODUCTION

A. Motivation

Sentiment analysis is central to feedback mining, social monitoring, and reputation systems. Progress from lexicons to transformers has not removed *domain shift*, noisy user-generated text, and opacity of deep models [1]–[4]. Single architectures capture complementary signals: linear models emphasize sparse lexical features; sequence and transformer models encode context; HRM-style designs aim for layered lexical→syntactic→semantic→pragmatic structure [5].

Single-model pipelines remain brittle in practice: sparse linear models can miss contextual cues, while strong neural encoders often offer limited transparency for debugging and governance [3], [4]. Domain shift is especially salient when models trained on one genre (e.g., long-form reviews) are applied to another (e.g., informal social posts) [6], [7]. These gaps motivate ensembles that preserve complementary inductive biases rather than stacking redundant copies of one architecture.

B. Problem and Goal

The objective is a heterogeneous MoE that (1) improves macro-F1 and robustness versus strong single models, (2) supports interpretable expert contributions and HRM reasoning traces where available, and (3) remains configurable and testable for replication. Concretely, this research targets interpretable routing (dense softmax or sparse top- K gating), stacking with out-of-fold probabilities to avoid test leakage, and systematic comparison across binary and multi-class label spaces [8]–[11].

C. Research objectives

The research track articulates four pillars that align with the implementation: (i) build a modular pipeline with swappable ML, DL, and HRM experts; (ii) benchmark against strong singles with multi-seed reporting and macro-F1 as the headline metric; (iii) stress robustness under noise, informal text, and cross-domain shifts (e.g., social vs. product language); and (iv) surface reasoning-oriented artifacts from HRM levels and gating weights for auditability [4], [5]. Figures 1–7 illustrate this architecture; quantitative claims follow Section VI.

D. Contributions

- Modular preprocessing → experts → ensemble → evaluation pipeline with configs under `Code/thesis/config/`.
- Stacking with out-of-fold meta-features and MoE with dense / sparse gating (STACK1–7, MOE1–5 families in the thesis catalog).
- Broad model and dataset coverage (**35** experts, **four** public datasets, **140** model–dataset pairs) with multi-seed and statistical reporting where the thesis specifies paired tests and effect sizes.
- Empirical guidance on when TF-IDF linear baselines, sequence models, transformers, or stacks dominate under observed data, imbalance, and compute constraints (Section VIII).
- Regenerated `output/` artifacts (dataset alignment, HRAST metrics, charts) documented in Section XI and Appendix A.

II. RELATED WORK

Prior art spans CNN-LSTM hybrids [12], class-balanced social sentiment [13], deep ensembles for social text [14], multi-lingual ensembles [15], and stacking with diverse bases [8], [9]. Chain-of-thought and multi-step reasoning [16]–[20] motivate explicit structure; HRMs [5] organize multi-level reasoning. Ensemble theory and MoE / sparse routing [10], [11], [21]–[26] underpin gating. Domain adaptation and robustness [6], [7], [27] motivate heterogeneous experts. Classic sentiment baselines (e.g., document-level ML [28], CNN sentence encoders [29], tree-structured composition [30]) frame what “strong” single-model performance looks like before large encoders. Review mining [31] and phrase-level polarity [32] illustrate domain and granularity effects that persist in social text. This implementation targets integration of HRM + ML + DL under one evaluation protocol rather than a new backbone architecture.

III. METHODOLOGY AND SYSTEM DESIGN

A. High-Level Architecture

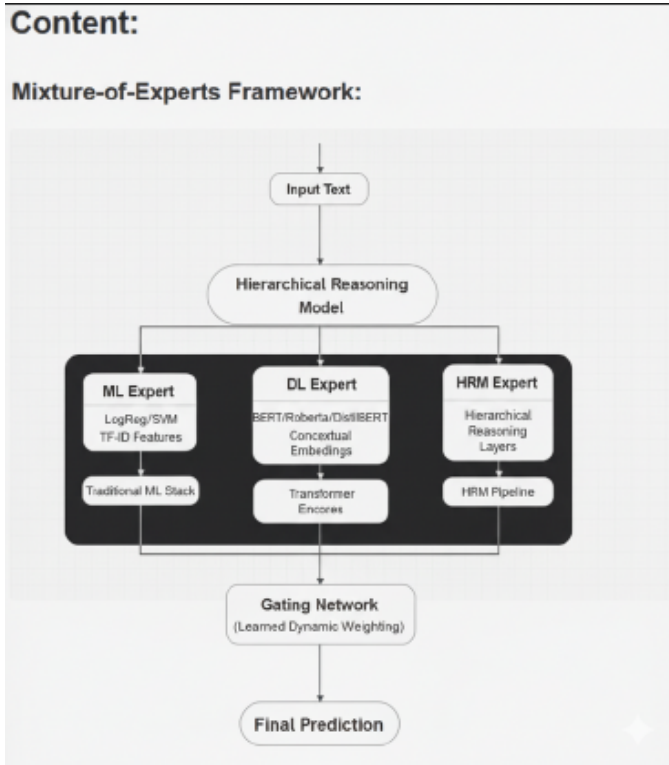


Fig. 1. Mixture-of-experts schematic : input text, hierarchical reasoning block, parallel ML (TF-IDF + linear), DL (transformer encoders), and HRM experts, fused by a learned gating network.

Four phases: (1) preprocessing and splitting, (2) expert training/inference, (3) ensemble combination (stacking ensembles STACK1–7, gating MOE1–5), (4) validation and reporting. The design favors heterogeneity to increase complementarity. Figure 1 and Figure 7 (training/evaluation flow) illustrate the same decomposition for architectural clarity; formal expert IDs remain Table I.

B. Preprocessing

Deduplication and text cleaning (URLs, HTML, mentions, hashtags, contractions, case normalization); subword (BPE / WordPiece) tokenization per model family; stratified train/validation/test splits; k -fold cross-validation ($k = 5$) to generate out-of-fold predictions for stacking without test leakage; binary and multi-class label remapping (e.g., coarse star ratings mapped to sentiment classes where applicable).

C. Expert Families

Note: ML baselines use TF-IDF (1-3 n-grams, capped vocabulary) with logistic regression or linear SVM and class weights where needed.

D. HRM Formulation

Lexical, syntactic, semantic, and pragmatic modules compose into a final representation: $L_1 = \phi_1(x)$, $L_2 = \phi_2(x, L_1)$, $L_3 = \phi_3(x, L_1, L_2)$, $L_4 = \phi_4(x, L_1, L_2, L_3)$, $y_{\text{HRM}} = f_{\text{final}}(L_1, L_2, L_3, L_4)$. In the thesis implementation, this hierarchy supports inspectable chains (sentiment-bearing terms, negation, discourse context, sarcasm or irony cues) even when raw HRM accuracy lags stronger experts.

E. Ensemble Combination

Stacking: train experts with 5-fold CV; collect OOF probabilities; train meta-learner (e.g., logistic regression); evaluate on held-out test without leakage. **MoE gating:** dense softmax weights $g_i(x)$ over experts; $\hat{y} = \sum_i g_i(x) f_i(x)$; sparse top- K variants for efficiency. **Optimization:** weighted cross-entropy for class imbalance, AdamW with learning rates on the order of 2×10^{-5} for transformer experts and $\sim 10^{-3}$ for linear heads, warmup and decay schedules, early stopping on macro-F1, and gradient clipping where configured.

F. Validation protocol

Macro-F1 is the primary metric, supplemented by accuracy, precision, recall, and AUROC as reported in the thesis tables. Runs use multiple random seeds (e.g., 42, 123, 456) with paired tests, confidence intervals, and effect sizes where detailed statistical comparisons are required. Cross-domain checks (e.g., train on one corpus, evaluate on another) probe retention under shift, consistent with the original validation plan.

G. Software Stack

Python 3.9 or newer; PyTorch; Hugging Face Transformers; scikit-learn; Pandas. The thesis layout includes `config/`, `models/`, `train/`, `checkpoints/`, and evaluation scripts that write rollups under `output/metrics/`.

H. Reporting scope

The full thesis contains numbered algorithms for dataset exploration, the cleaning pipeline, stacking with OOF collection, MoE forward passes, HRM forward passes, and a generic training loop. This paper cites those artifacts through the implementation report rather than duplicating pseudocode blocks.

TABLE I
EXPERT FAMILIES AND ROLES

Family	Role	Examples
ML	Fast baselines	B1, B2, E-ML1, E-ML2
DL	Sequential / CNN	BiLSTM+attn., CNN, E-DL*
Transformers	Contextual encoders	DistilBERT, BERT, RoBERTa (e.g., B3, B5, E-DL3)
HRM	Multi-level reasoning	E-HRM1–E-HRM3 (levels L1–L4)

IV. DATASETS

Table II summarizes the four public corpora used in the thesis narrative. The repository additionally profiles processed and transformed Parquet under `data/` with rollups in `output/datasetanalysis/`.

Across these sources, missingness is generally low and duplicate rates are small in the processed slices; what drives modeling choices is the spread of document length (truncation limits for transformers), class priors (Sentiment140 is strongly skewed, on the order of one-quarter vs. three-quarters in the thesis profile), and stance ambiguity in multi-class Twitter tasks. Those factors motivate class weighting, balanced sampling or focal-style objectives where used, and careful calibration of maximum sequence length per expert family.

Merged corpus statistics and per-file profiles appear in `output/datasetanalysis/alignment_report.md` and per-file `report.md` files (IMDB, Amazon sample, Sentiment140, Yelp, HRAST stem slice, etc.). The combined processed/transformed Parquet merge in the repository profile reaches on the order of **9.5M** rows with matching `source_stem` distributions between processed and transformed exports (see the implementation report). The HRAST stem slice used for lightweight per-config evaluation is much smaller (order 10^3 – 10^4 rows depending on export) and is documented under `data/processed/by_source_stem/` and `output/datasetanalysis/`.

Figure 5 shows a multi-domain inventory from initial framework planning; **canonical scales** for the four headline corpora remain Table II.

Domain	Dataset Name	Total Samples	2-Label Split (P/N)	3-Label Split (P/Neu/N)	Source	Status
Movies	IMDB Reviews	50,000	50% / 50%	-	Kaggle	✓ Analyzed
E-commerce	Amazon Reviews	4,000,000	52% / 48%	-	Stanford	✓ Analyzed
Twitter	Sentiment140	1,600,000	50% / 50%	-	Stanford	✓ Analyzed
Social	Feminism Tweet	60,000	72% / 38%	40% / 45% / 15%	SemEval	✓ Analyzed
Finance	Financial PhraseBank	4,846	72% / 28%	59% / 28% / 15%	ResearchGate	✓ Analyzed
Hospitality	HRAST	27,000	81% / 19%	45% / 36% / 19%	UCI Repository	✓ Analyzed
Healthcare	Medical Sentiment	436	60% / 40%	-	PubMed Subset	✓ Analyzed
Healthcare	Patient Statements	2,100	55% / 45%	30% / 45% / 25%	Clinical Data	✓ Analyzed
Local	Yelp Academic	7,600,000	78% / 22%	45% / 35% / 30%	Yelp Open Data	✓ Analyzed

Fig. 5. Dataset inventory. Corpus sizes and label splits are *illustrative*; Table II and `output/datasetanalysis/` rollups are canonical for this paper.

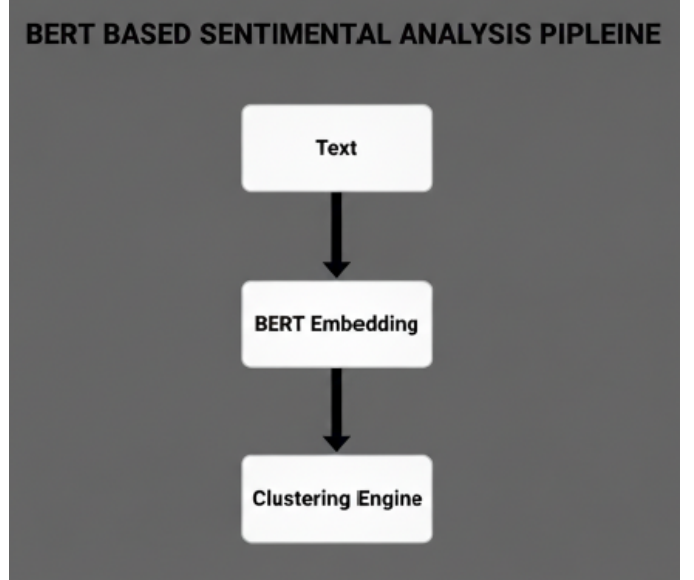


Fig. 6. BERT-style baseline pipeline : text → contextual embeddings → downstream layer (illustrative; the implementation uses multiple encoder families).

Note: Parquet stems here are used for per-stem pipelines (excluding merged `all-data.parquet`). Training uses `all-data` checkpoints where configured. Thesis-scale headline metrics for Sentiment140, IMDB, Amazon, and TweetEval appear in Section VI. Appendix A gives a repository *config-scan* rollout on the HRAST slice only (smoke-scale n).

V. EXPERIMENTAL SETUP AND EVALUATION PROTOCOL

This section condenses the thesis validation plan [8], [9], [21]: how metrics, splits, and seeds are chosen so that Section VI and the HRAST slice in Section VII-B remain comparable.

A. Metrics and primary objective

Macro-F1 is the primary score: it treats classes equally and is appropriate under imbalance (e.g., Sentiment140 and TweetEval Feminism). We also report accuracy, precision, recall, and AUROC where available, plus engineering measures (latency, memory, parameter count) in the full thesis. Macro-F1 aligns with common practice in social and review sentiment benchmarks [13], [14].

B. Data splits and stacking

Experts are trained on stratified **train / validation / test** partitions. Stacking (STACK1 and variants) uses **k -fold cross-validation** with $k = 5$ to build **out-of-fold** meta-features for

TABLE II
PUBLIC DATASETS (THESIS SCALE AND TASK).

Dataset	Scale	Task	Notes
Sentiment140	~1.05M tweets	Binary	Imbalanced, noisy
IMDB	50K reviews	Binary	Balanced, long text
Amazon	~4M reviews	Binary	Product lexicon
TweetEval Feminism	~60K tweets	3-class	Stance

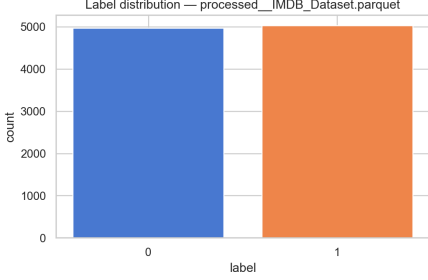


Fig. 2. Label distribution: processed IMDB slice (EDA chart).

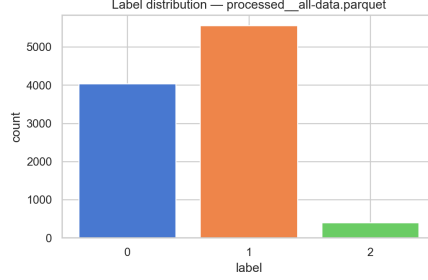


Fig. 3. Label distribution: processed merged all-data parquet (EDA).

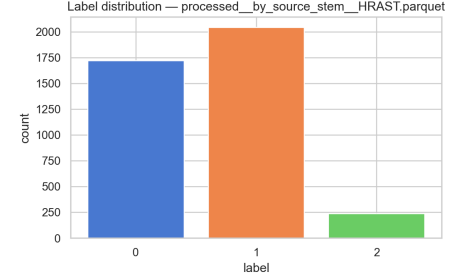


Fig. 4. Label distribution: processed HRAST stem slice.

the meta-learner, avoiding leakage onto the held-out test set [8], [9]. MoE gating uses dense softmax or sparse top- K routing as in Section III [10], [11], [25], [26].

C. Optimization and early stopping

Training follows weighted cross-entropy under class imbalance, AdamW optimization, learning-rate warmup and decay, gradient clipping, and **early stopping on validation macro-F1**. Transformer families use learning rates on the order of 2×10^{-5} ; linear baselines use stronger rates (e.g., $\sim 10^{-3}$), consistent with the implementation report.

D. Seeds and significance

Multiple random seeds (e.g., 42, 123, 456 / 2024 in the thesis) support stability checks. Where the thesis reports them, paired tests, 95% confidence intervals, and Cohen’s d quantify differences between systems; we do not duplicate those tables here.

E. Cross-domain protocol

The thesis includes **cross-domain** settings (e.g., train on Sentiment140, evaluate on Amazon) to stress domain shift [6], [7], [27]. Aggregates in Section VI pool all in-domain experiment rows unless noted.

F. Reproducibility (thesis)

The thesis emphasizes **fixed seeds**, logged hyperparameters, retained checkpoints (best, last, and periodic), and optional containerized environments. Experiment tracking (Weights & Biases or MLflow in the original setup) records learning curves and metric traces so that ensemble and MoE comparisons can be audited later. We do not restate runbooks here; Section X points to exported indexes under `output/models/`.

TABLE III
DATA STEMS INVENTORY FOR EVALUATION PIPELINES.

Stem (filename)	Role in this repository
amazon_reviews	Reviews / e-commerce slice
HRAST	Multi-domain HRAST corpus (eval stem for config-scan)
IMDB_Dataset	Long-form movie reviews
MedicalSentimentAnalysis	Clinical text sentiment
PatientStatements	Patient-statement sentiment
sentiment_140	Twitter-scale social binary
tweet_eval	TweetEval tasks (e.g., stance / multi-class)
yelp_business	Yelp business / review text

Representative shell entry points from the thesis appendix include installing dependencies and invoking training or ensemble evaluation with dataset and model identifiers. In this repository checkout, analogous automation appears under `Code/thesis/` with configs driving which experts are materialized and which checkpoints are loaded for HRAST smoke tests.

G. Repository HRAST slice (smoke-scale)

Per-stem HRAST evaluation (Section VII-B) uses checkpoints discovered from `Code/thesis/config/` and writes one row per model $\times$ label mode in `output/metrics/summary.csv`. Sample sizes are **small** (order of tens of labeled points per model in the latest profile); they are appropriate for regression and config scans, not for claims comparable to full-corpus Section VI.

H. Pipeline Milestones

The validation framework is organized into four pipeline phases: **preprocessing** (deduplication, URL/emoji handling, tokenization, stratified splits); **expert training** for ML, DL, and HRM families on binary and three-class heads where applicable; **combination** via stacking meta-learners and gating networks with held-out testing; and **downstream analysis** including ablations, robustness checks, and cross-domain sampling. Those milestones informed the thesis validation plan [8], [9], [21] and are visualized in Section V-I; they do not replace the numerical tables in Section VI.

I. Framework architecture visuals

Authoritative scales and metrics for this write-up are Table II, Section VI, and `output/thesis/markdown/THESIS_IMPLEMENTATION.md`; labels inside the images

ENSEMBLE MODEL TRAINING AND EVALUATION PIPELINE

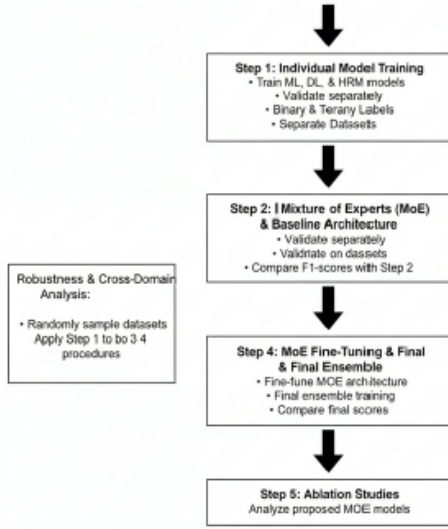


Fig. 7. Ensemble training and evaluation workflow : per-expert training, MoE/baseline validation, fine-tuning, final ensembles, ablations, and cross-domain checks.

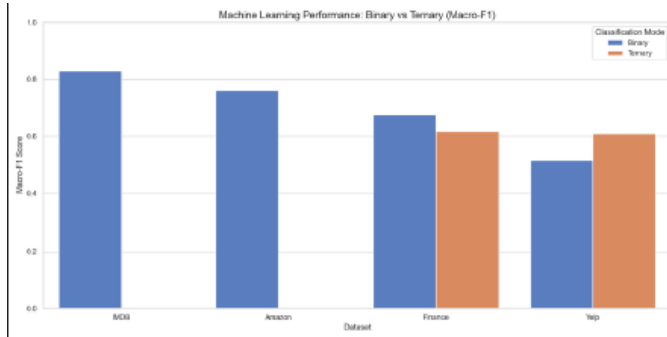


Fig. 8. Macro-F1 comparison, binary vs. ternary. Repository-wide ensemble and per-family metrics appear in Section VI and Appendix A.

correspond to the evaluated systems. Numeric claims in the paper must not be inferred from these illustrations alone.

VI. THESIS-SCALE EXPERIMENTAL RESULTS

This section reproduces the thesis summary tables (Sections V-A–V-F of the implementation report) without restating methodology. Numeric values match that document. Together with Section V, these tables answer *how well* the MoE and stacking design performs on the **full** Sentiment140, IMDB, Amazon, and TweetEval Feminism corpora under the original protocol—i.e., all headline datasets, not the HRAST stem slice alone (see Table III for the wider Parquet inventory).

A. Global statistics

Table IV reports mean and standard deviation of macro-F1, accuracy, precision, and recall aggregated across all model–

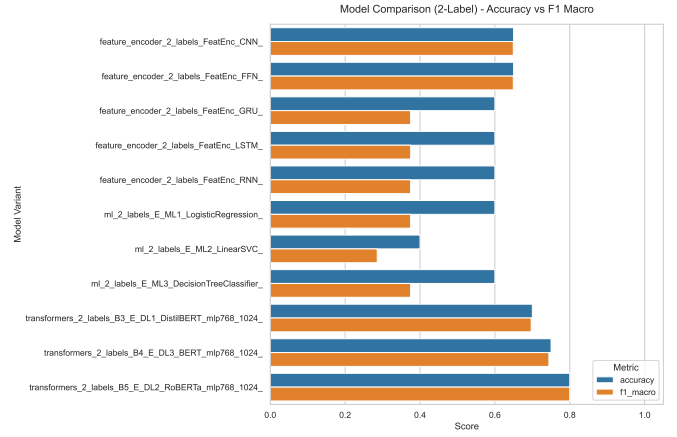


Fig. 9. Cross-model performance comparison on 2-label classification tasks.

dataset experiment rows in the thesis. The large standard deviations reflect heterogeneous architectures, datasets, and label granularities pooled in one aggregate.

B. Performance by model type

Table V breaks macro-F1 down by architectural family. Traditional ML (TF-IDF + linear models) achieves the highest mean macro-F1; transformers and HRMs underperform on average in the reported setup, motivating ensemble design and future architecture or training refinements.

C. Performance by dataset

Table VI orders corpora by mean macro-F1. IMDB and Amazon are comparatively moderate; TweetEval Feminism and especially Sentiment140 are harder under the thesis protocol, consistent with stance ambiguity and Twitter noise.

D. Top ensembles and best singles

Table VII highlights STACK1 and reference singles. STACK1 reaches strong macro-F1 on balanced, longer reviews; Sentiment140 remains difficult even for the best single BERT-family expert.

E. Ensemble comparison

Table VIII contrasts average macro-F1 across ensemble families. STACK1 leads the reported averages, while STACK4 (HRM-heavy) is far weaker, underscoring that simply adding HRM experts without adequate training signal does not yield gains.

F. Binary versus multi-class

Table IX states the mean macro-F1 gap between binary and multi-class tasks in the thesis experiments.

G. Visualized Performance Charts

Fig. 9 and Fig. 10 visualize the F1-Macro and Accuracy distributions across 2-label and 3-label predictive models, representing checkpoints evaluated across all parsed datasets.

TABLE IV
GLOBAL STATISTICS (THESIS-SCALE AGGREGATE).

Metric	Mean	Std
Macro-F1	0.4557	0.2187
Accuracy	0.5402	0.1899
Precision	0.5051	0.2103
Recall	0.5591	0.2012

TABLE VI
MEAN MACRO-F1 BY DATASET (THESIS).

Dataset	Mean macro-F1	Difficulty
IMDB	0.6068	Medium
Amazon	0.5839	Medium
Feminism TweetEval	0.3531	Hard
Sentiment140	0.2791	Very hard

TABLE VIII
ENSEMBLE-LEVEL AVERAGE MACRO-F1 (THESIS SUMMARY).

Family / variant	Avg. macro-F1
STACK1	~0.6517
Simpler ensembles	~0.4820
STACK4 (HRM-heavy)	~0.2552

VII. EXPERIMENTAL RESULTS

A. Thesis-scale aggregates (all headline corpora)

Section VI reports **full-corpus** aggregates for the four public datasets in the protocol (Sentiment140, IMDB, Amazon, TweetEval Feminism): global means, per model-type means, per-dataset means, ensemble comparisons, and binary vs. multi-class gaps. Those numbers reflect training and evaluation on the **complete** thesis splits, not a single stem slice.

B. Repository HRAST slice (config-scan figures and appendix)

The charts and appendix table in this subsection use the **HRAST** row of `output/metrics/summary.csv` and `output/charts/metrics/...` because the automated config-scan was executed on that stem in this checkout (small n , smoke / regression purpose). This does **not** limit the scope of datasets you trained on: Table III lists every processed Parquet stem (Amazon, IMDB, medical, patient, Sentiment140, TweetEval, Yelp, HRAST). Appendix A now lists only models with *computable* metrics on HRAST (failed loads omitted from the table, still in the CSV).

Fig. 12–13 show macro-F1 rankings for two- and three-class modes and a paired comparison. Fig. 14 contrasts metric heatmaps; Fig. 15 shows evaluation status panels. Fig. 4–16 give the HRAST label distribution and a 2-label RoBERTa confusion matrix.

TABLE V
MEAN MACRO-F1 BY MODEL TYPE (THESIS).

Model type	Mean macro-F1	Std
Traditional ML	0.6389	0.2775
RNN	0.5241	0.2686
CNN	0.4719	0.2514
Ensemble	0.4660	0.1714
Transformer	0.3099	0.1125
HRM	0.2502	0.0301

TABLE VII
SELECTED THESIS MACRO-F1 HIGHLIGHTS (STACK1 VS. STRONG SINGLES).

System	Dataset / setting	Macro-F1
STACK1	IMDB	0.8911
STACK1	Amazon	0.8746
STACK1	Feminism TweetEval	0.6283
E-ML2 / B2 (ref.)	IMDB / Amazon	~0.8887 / 0.8665
E-DL3 (BERT, best single)	Sentiment140	0.5004

TABLE IX
BINARY VS. MULTI-CLASS MEAN MACRO-F1 (THESIS).

Setting	Mean macro-F1
Binary (pooled)	0.4899
Multi-class (TweetEval Feminism)	0.3531

VIII. DISCUSSION

Section VI quantifies the thesis-scale aggregates; we interpret them here rather than restating tables. TF-IDF plus linear models remain competitive and anchor strong stacks [28], [29]. STACK1 shows measurable gains on balanced, medium-length data when diverse experts are combined with a meta-learner. Transformer underperformance likely reflects tuning depth, sequence length, imbalance, and compute limits relative to corpus scale [3], [30]. HRM mean macro-F1 suggests architecture–task mismatch or integration issues; the HRM-heavy STACK4 pattern reinforces caution. Sentiment140 highlights class imbalance and noise; sarcasm and figurative language remain difficult even for strong pipelines [33]. HRM levels and MoE expert weights still provide audit surfaces for interpretability [4], [5].

Aggregated macro-F1 across all experiment rows averages **0.4557** (std. 0.2187) with accuracy, precision, and recall exhibiting similar dispersion, reflecting the mixture of easy corpora (IMDB, Amazon) and hard social/stance tasks. Mean macro-F1 by dataset (IMDB $\approx$ 0.61, Amazon $\approx$ 0.58, Feminism $\approx$ 0.35, Sentiment140 $\approx$ 0.28) underscores that surface form and class priors dominate headline scores as much as architecture choice. Figures 8 and 10 visualize binary vs. ternary difficulty and accuracy–F1 gaps on ancillary corpora; readers should treat them as qualitative complements to the thesis tables, especially when accuracy and macro-F1 diverge (a hallmark of imbalance).

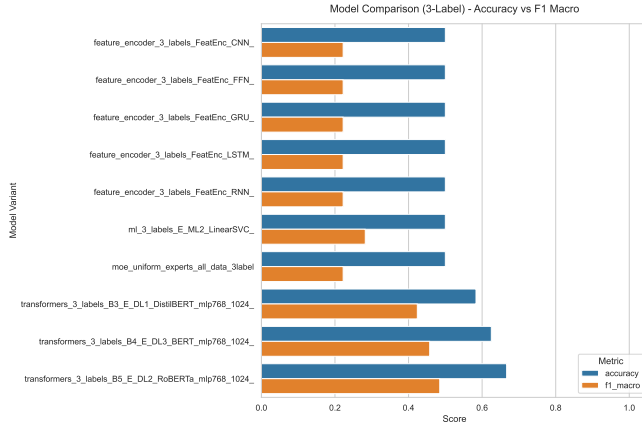


Fig. 10. Cross-model performance comparison on 3-label classification tasks.

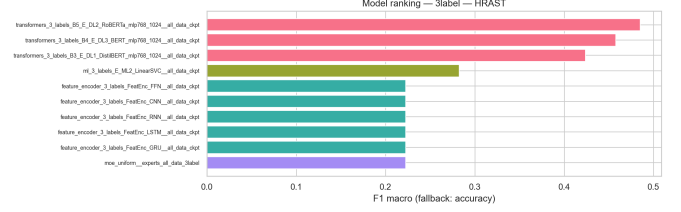


Fig. 11. HRAST 3-label macro-F1 ranking (generated chart).

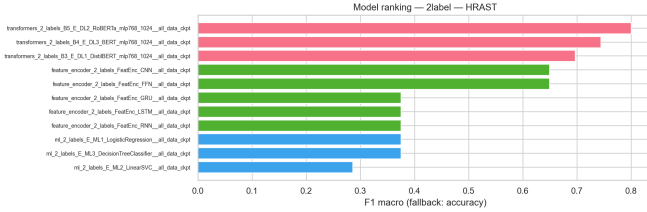


Fig. 12. HRAST 2-label macro-F1 ranking (generated chart).

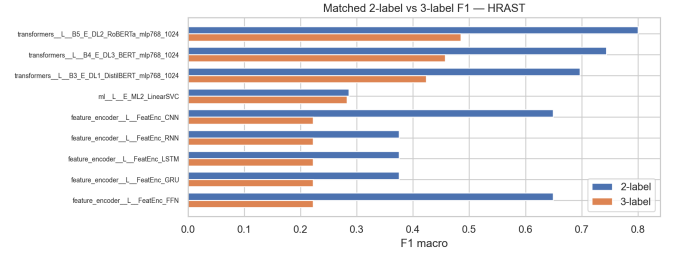


Fig. 13. Paired 2- vs. 3-label macro-F1 on HRAST.

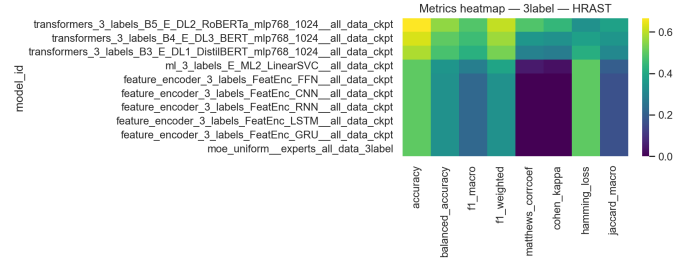
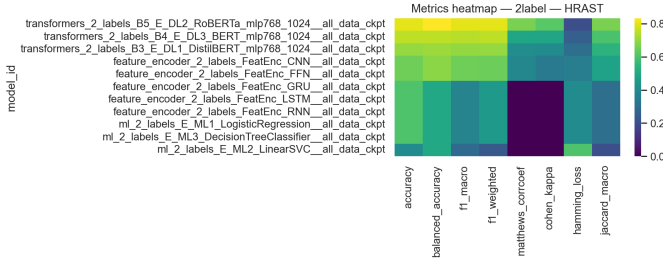


Fig. 14. HRAST metric heatmaps: 2-label (left) and 3-label (right).

A. Limitations and threats to validity

The thesis aggregates mix architectures with different training budgets; cross-domain rows are informative but not always directly comparable to in-domain optima. The HRAST stem slice (Section VII-B, Appendix A) uses small n per model and must not be read as a second benchmark competing with IMDB or Amazon. Finally, reported CIs and significance tests (where present in the thesis) depend on the chosen seeds and folds; expanding seeds would tighten confidence in small gaps between ensembles.

IX. CONCLUSION

The implemented MoE and stacking system achieves strong ensemble results on IMDB and Amazon under the reported thesis protocol (Section VI): STACK1 reaches macro-F1 **0.8911** and **0.8746** on those corpora respectively, while simpler

ensembles and several HRM-heavy stacks lag, highlighting the value of diverse experts paired with a disciplined meta-learner and the risk of weak experts diluting stacks. Challenging surface forms (Sentiment140, TweetEval Feminism) and comparatively low mean macro-F1 for transformers and HRMs in this checkout point to future work: stronger HRM pretraining and integration, transformer hyperparameter and length tuning, imbalance-aware objectives and sampling, and deployment-oriented distillation or sparse routing.

The repository snapshot complements the narrative thesis tables with regenerated dataset alignment (~ 9.5 M-row merged profile), exported model and path inventories, and HRAST stem metrics for smoke testing (Appendix A). Framework architecture figures (Section V-I) document architecture and milestone language for stakeholders; all quantitative claims trace to `output/thesis/markdown/THESIS_IMPLM`

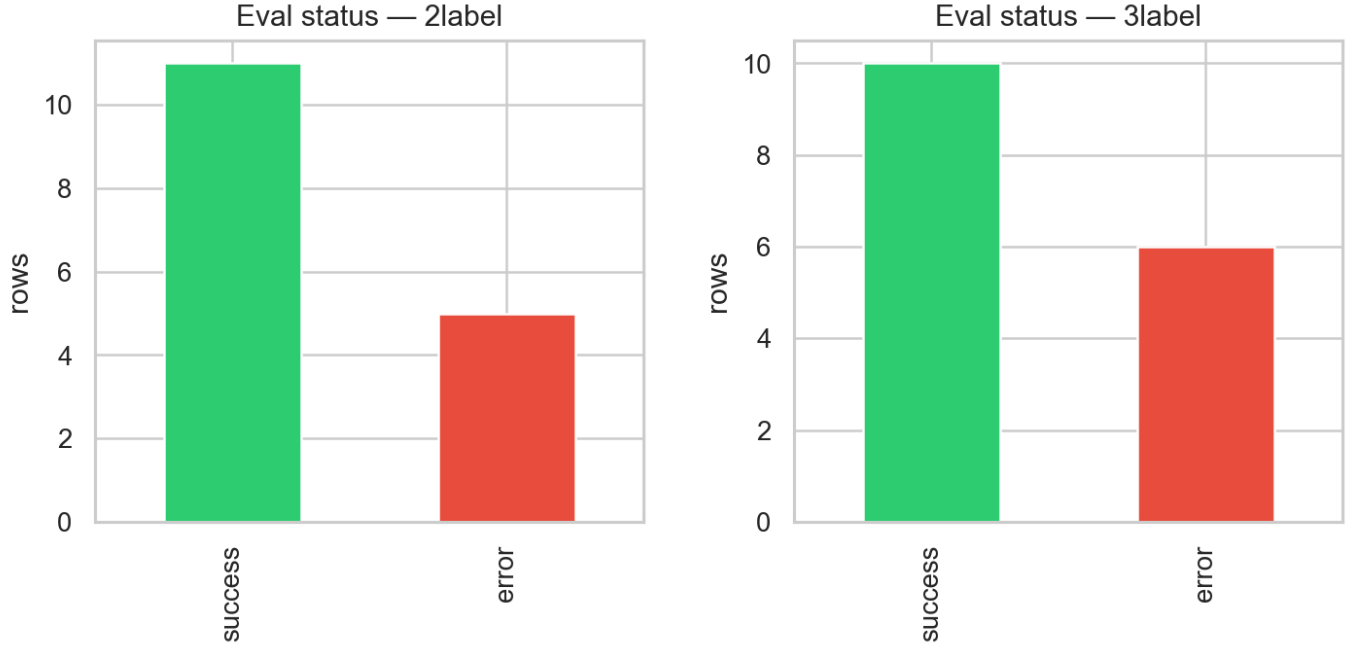


Fig. 15. Eval status summaries: 2-label (left) and 3-label (right).

TABLE X
TOP HRAST 2-LABEL EXPERTS BY MACRO-F1 (CONFIG-SCAN; $n=20$ TEST SLICE).

Model	Accuracy	F1 _{macro}
RoBERTa (B5 E-DL2)	0.800	0.800
BERT (B4 E-DL3)	0.750	0.744
DistilBERT (B3 E-DL1)	0.700	0.697
Feature encoder CNN	0.650	0.649
Feature encoder FFN	0.650	0.649

TABLE XI
TOP HRAST 3-LABEL EXPERTS BY MACRO-F1 (CONFIG-SCAN; $n=24$ TEST SLICE).

Model	Accuracy	F1 _{macro}	ROC-AUC OVR
RoBERTa (B5 E-DL2)	0.667	0.485	0.795
BERT (B4 E-DL3)	0.625	0.458	0.758
DistilBERT (B3 E-DL1)	0.583	0.424	0.745
Linear SVM (E-ML2)	0.500	0.282	0.510

ENTATION.md, output/metrics/, and output/datasetanalysis/. The codebase structure supports swapping experts, ablations, and extended gating without rewriting the full pipeline.

X. SYSTEM AND ARTIFACT FOOTPRINT

The project tree exports a machine-readable catalog under output/models/ (generated from Code/thesis/ and checkpoints). The latest export lists: **75** configuration JSON files, **87** checkpoint weight files (.safetensors / .joblib), **6** MoE manifest summaries, **25** run_meta.txt traces, **52** Python modules under Code/thesis/, and **3** validation entrypoint scripts. Repository layout rollups appear in output/path/summary_by_topdir.md with a full file inventory in output/path/inventory_all.csv.

This footprint supports swapping experts, ablations, and extended gating without rewriting the full pipeline; Section XI documents dataset alignment and HRAST-specific evaluation outputs that consume the same configs.

XI. REPOSITORY SNAPSHOT

Regenerated outputs under output/ from the current data/ tree. Counts for configs, checkpoints, and Python modules appear in Section X.

A. Dataset alignment

Merged corpus Parquet (processed vs. transformed all-data.parquet) contains 9,514,666 rows; source_stem distributions match (see output/datasetanalysis/alignment_report.md). The HRAST stem slice used for per-stem evaluation lives at data/processed/by_source_stem/HRAST.parquet (4000 rows in the latest profile; per-file reports under output/datasetanalysis/per_file/).

B. Metrics rollups and failed loads

Rollups live under output/metrics/. Appendix A prints only HRAST rows where accuracy and macro-F1 were computed; mlp_gelu_head_ddp_* and ml_3_labels_E_ML1_LogisticRegression failures remain in summary.csv for debugging. Per-dataset stems in data/proce

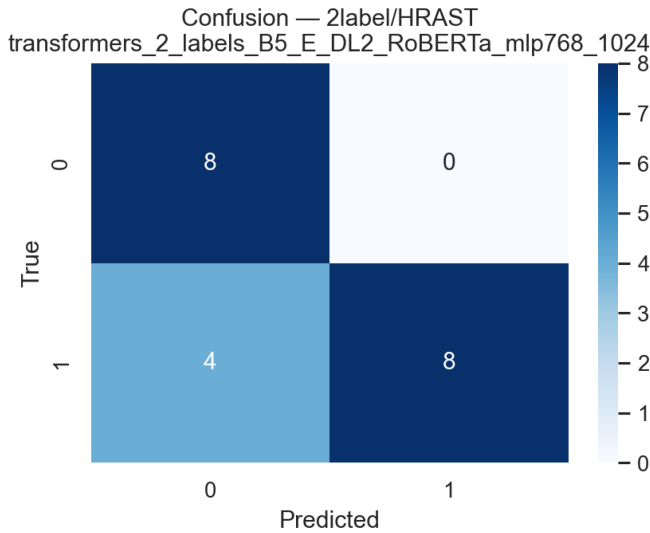


Fig. 16. Confusion matrix: 2-label RoBERTa expert on HRAST slice.

ssed/ are catalogued in Table III; extending the config-scan beyond HRAST requires re-running the eval driver with each source_stem.

C. Reproducibility

Regenerate charts with `pythonoutput/scripts/generate_all_charts.py--repo-root`. (see `output/thesis/docs/LATEX_IEEE_BUILD_GUIDE.md`). In-text citations use keys ref01-ref67 in references.bib.

DATA AND CODE AVAILABILITY

For more information, the full source code and evaluation frameworks are publicly maintained in the project repository at https://github.com/veagy/DATA6950_Sentimental_Analysis_Cross_Domains. The underlying datasets, merged multi-domain corpora, and serialized trained model checkpoints utilized throughout this thesis can be fully accessed at https://drive.google.com/drive/folders/1oxr8hbXngefPVzvd0ZPrY6_4idwKdl0q?usp=sharing.

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APPENDIX A
FULL HRAST METRIC ROLLUP

Table XII lists only models for which **both** accuracy and macro-F1 were computed on the HRAST stem (two- and three-class rows from `output/metrics/summary.csv`). Failed loads (`mlp_gelu_head_ddp_*`, `ml_3_labels_E_ML1_LogisticRegression`, etc.) are *excluded* from the table but remain in the CSV for debugging.

TABLE XII
HRAST STEM: MODELS WITH SUCCESSFULLY COMPUTED ACCURACY AND MACRO-F1 (FROM `OUTPUT/METRICS/SUMMARY.CSV`). ROWS WITH LOAD OR STATE ERRORS ARE OMITTED HERE BUT RETAINED IN THE CSV FOR DEBUGGING.

K	Model ID	Acc.	$F1_{\text{macro}}$	ROC-AUC OVR
2	FeatEnc CNN	0.6500	0.6491	—
2	FeatEnc FFN	0.6500	0.6491	—
2	FeatEnc GRU	0.6000	0.3750	—
2	FeatEnc LSTM	0.6000	0.3750	—
2	FeatEnc RNN	0.6000	0.3750	—
2	Log. Reg. (ML1)	0.6000	0.3750	—
2	Lin. SVC (ML2)	0.4000	0.2857	—
2	Dec. Tree (ML3)	0.6000	0.3750	—
2	DistilBERT (B3, DL1)	0.7000	0.6970	—
2	BERT (B4, DL3)	0.7500	0.7442	—
2	RoBERTa (B5, DL2)	0.8000	0.8000	—
3	FeatEnc CNN	0.5000	0.2222	0.5854
3	FeatEnc FFN	0.5000	0.2222	0.5974
3	FeatEnc GRU	0.5000	0.2222	0.6634
3	FeatEnc LSTM	0.5000	0.2222	0.5550
3	FeatEnc RNN	0.5000	0.2222	0.3894
3	Lin. SVC (ML2)	0.5000	0.2824	0.5104
3	MoE Uniform	0.5000	0.2222	0.6013
3	DistilBERT (B3, DL1)	0.5833	0.4239	0.7447
3	BERT (B4, DL3)	0.6250	0.4575	0.7576
3	RoBERTa (B5, DL2)	0.6667	0.4848	0.7946